

FMTS mätteknik HT2026 – Dag 2

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Struan Gray

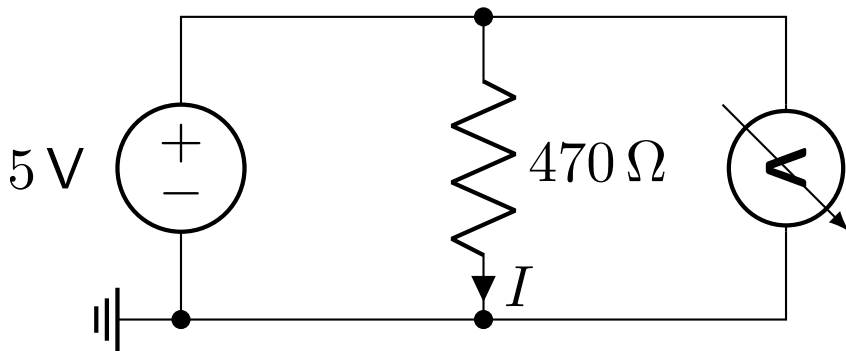
Akademien för informationsteknologi, Högskolan i Halmstad

September 21, 2026

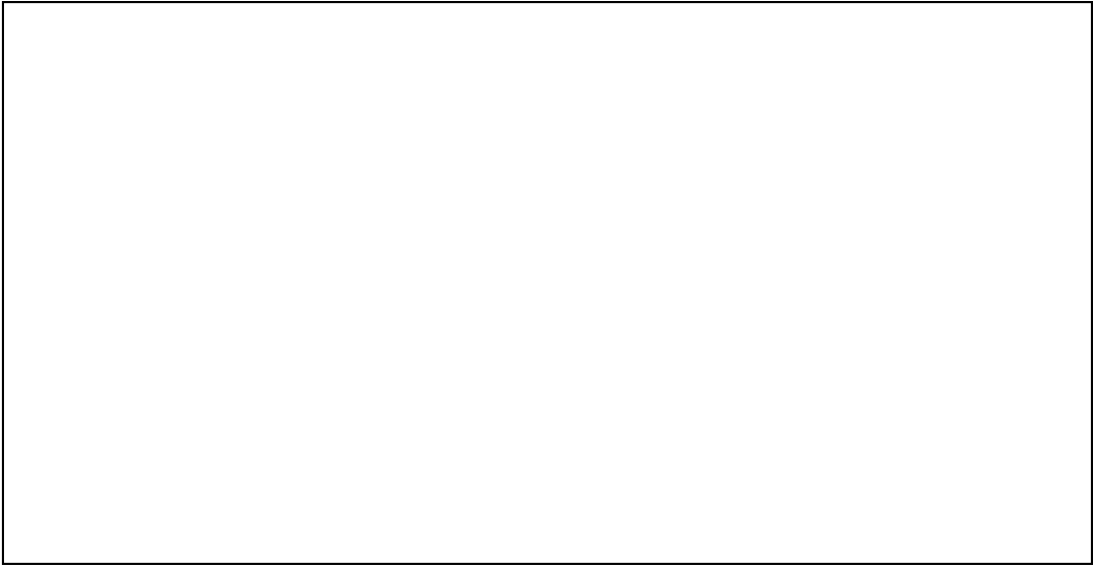
Dagens plan

- ▶ Repetera Ohms lag, mäta ström och spänning
- ▶ Öva spänningsdelning och kontrollmäte
- ▶ Närma oss gränsen för multimeteren
- ▶ AC - växelström
- ▶ RMS vs. True RMS

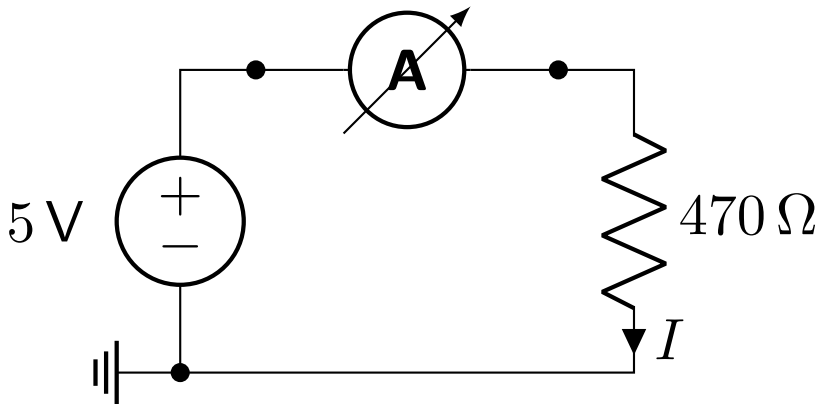
Mäta spänning



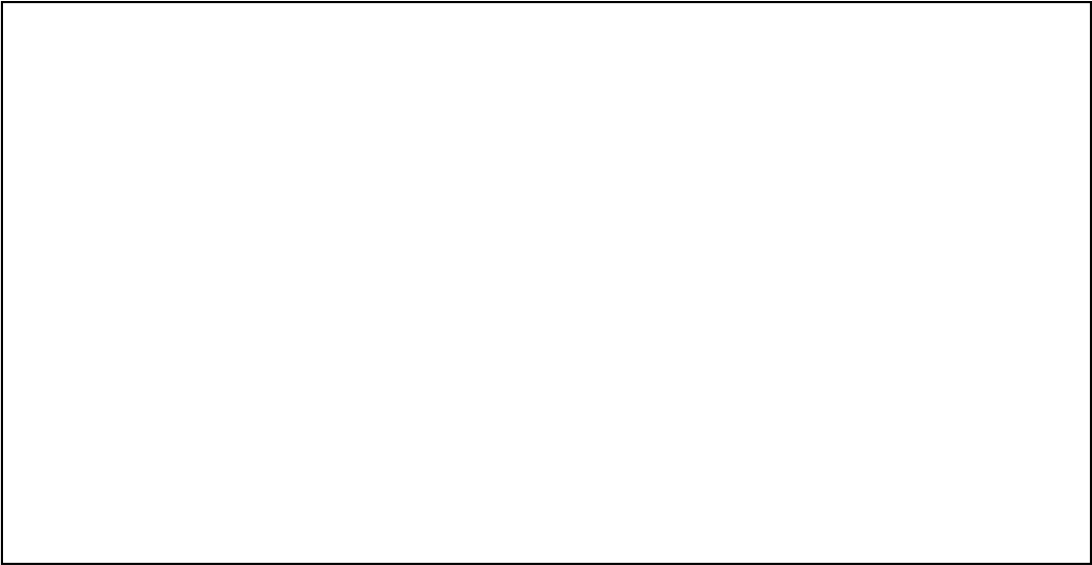
Spänning **över** komponenter!



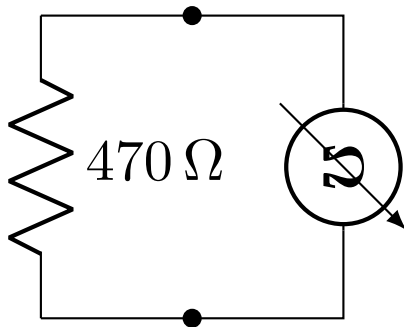
Mäta ström



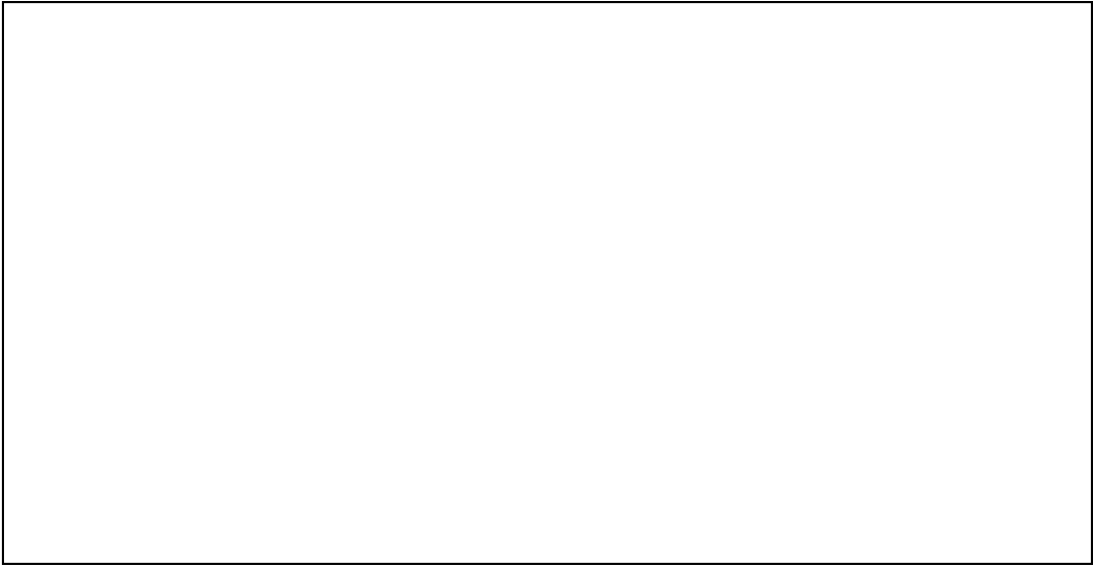
Ström **genom** komponenter!



Mäta motstånd



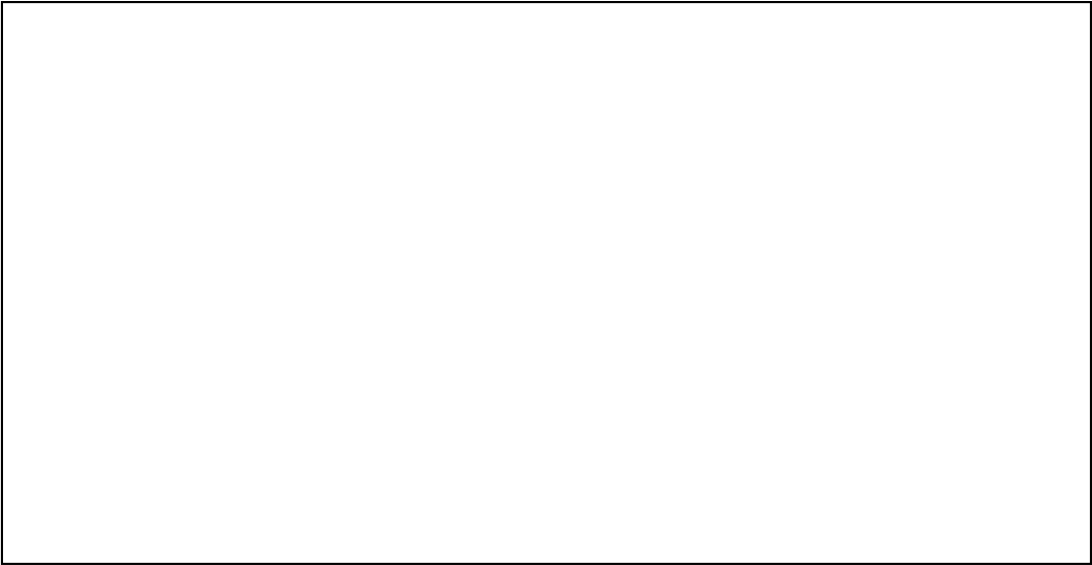
Koppla loss komponenter!



PM 2505 electr VA Ω meter PHILIPS

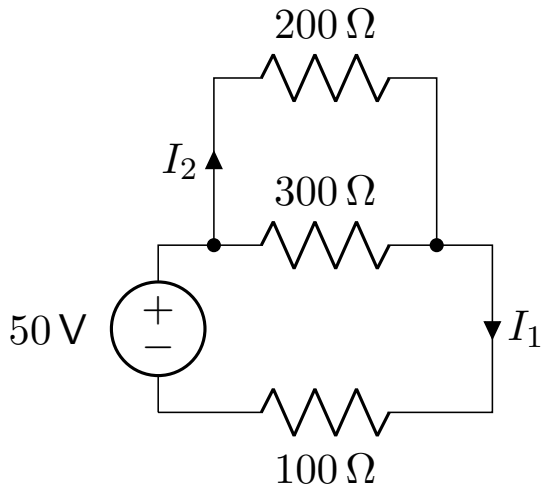
Klurigt!

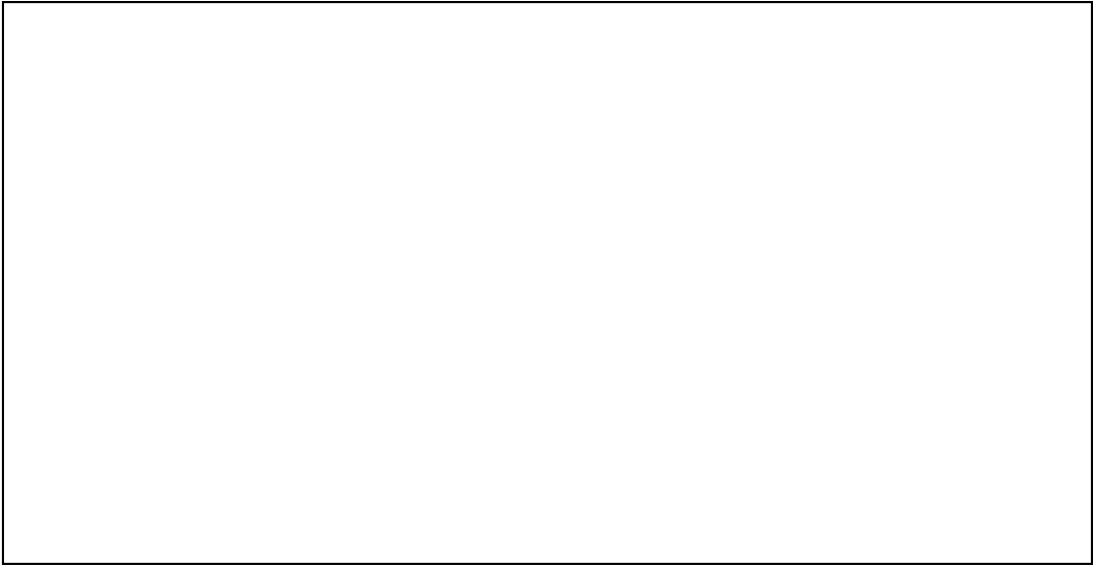




Ohms lag-repetition

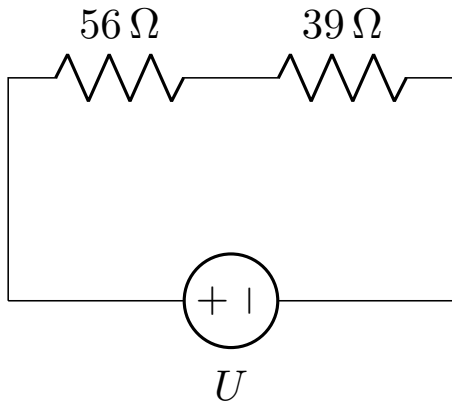
Beräkna I_1 och I_2 .

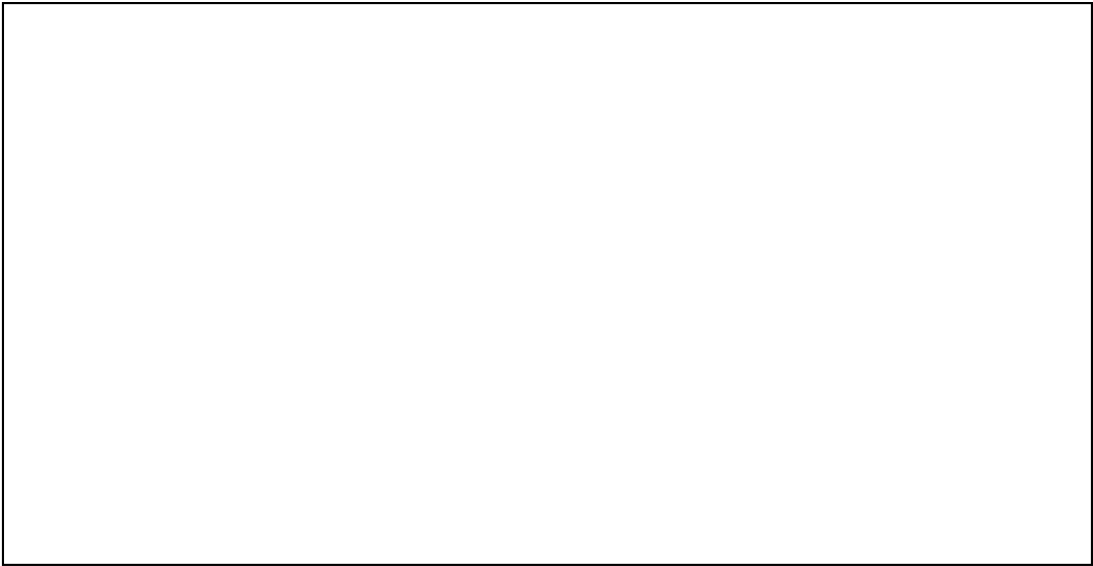




Effektbegränsning

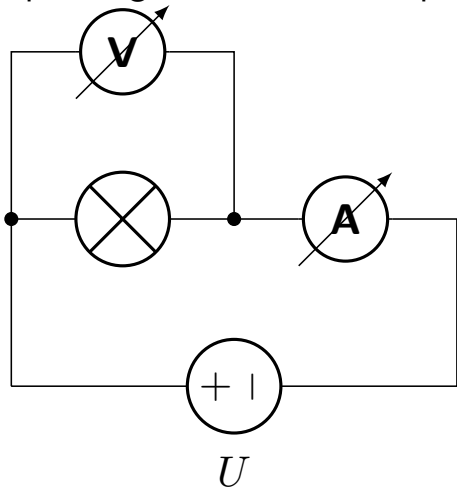
Två 1/4-wattsmotstånd på $56\ \Omega$ och $39\ \Omega$ kopplas i serie till en spänning U . Beräkna hur stor U högst får vara utan att något av motstånden överbelastas.

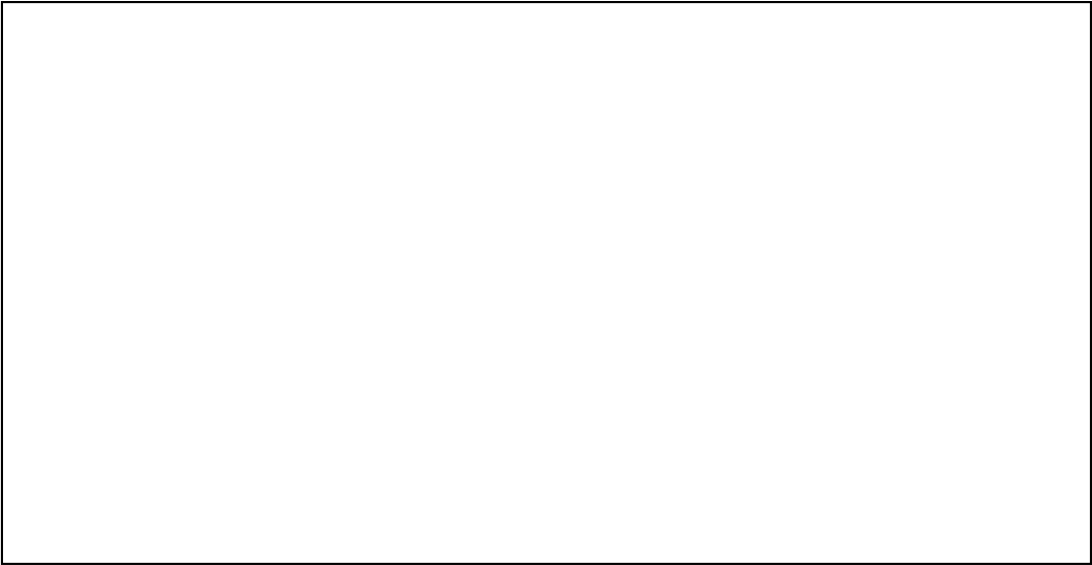




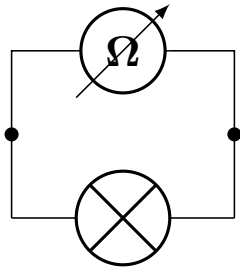
Mätrepetition

Koppla upp kretsen nedan (krysset är en lampa). Källspänningen U skall sättas till det som står på lampan (4 V eller 8 V typiskt). Mät ström och spänning och beräkna lampans resistans.

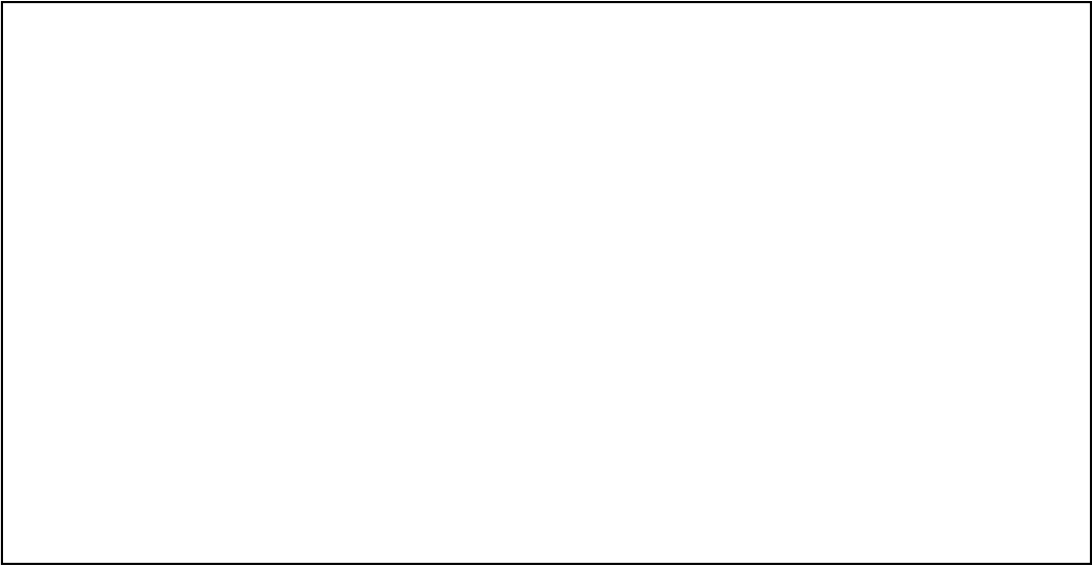


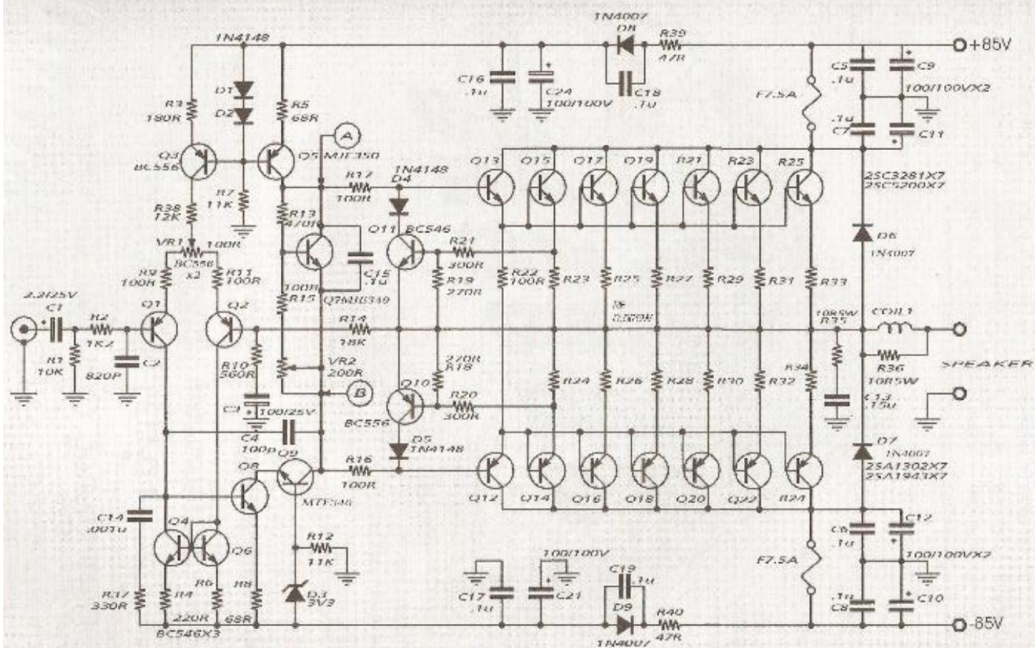


Koppla *ur* lampan och mät dess resistans med Multimetern.



Skall vara ungefär samma?





Right Channel
Speaker



Amplifier circuit protected by
Hegel US Patent No 6.275.104

H390 Integrated Amplifier
Hegel, Oslo, Norway

Serial Number



Apparatet må tilkoples
jordet stikkontakt

230VAC Fuse: 6.3AT

Line Level Outputs

Fixed

Variable

Left



Right



Analog Inputs

Right

XLR

Left

Analog 1

Analog 2

Left



Right



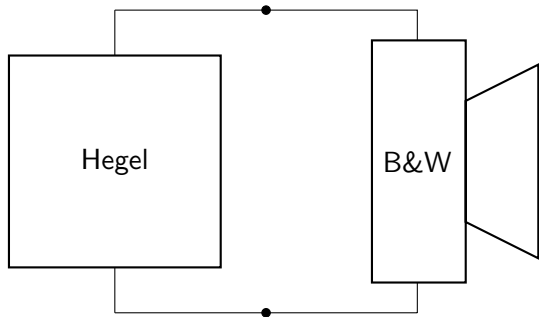
Left Channel
Speaker



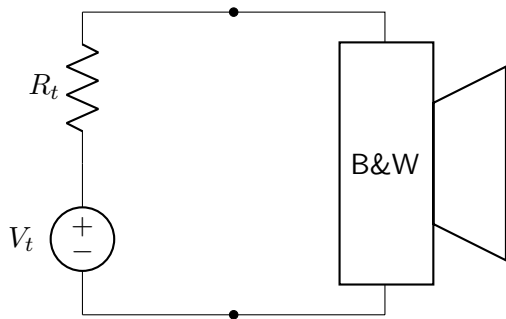
Digital
Out



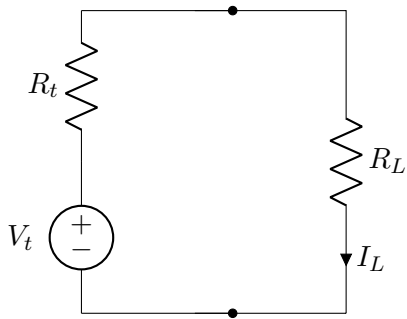
Effektanpassning – spänningsdelning



Effektanpassning – spänningsdelning



Effektanpassning – spänningsdelning



▶ Effekt i R_L är $P_L = U_L I_L$

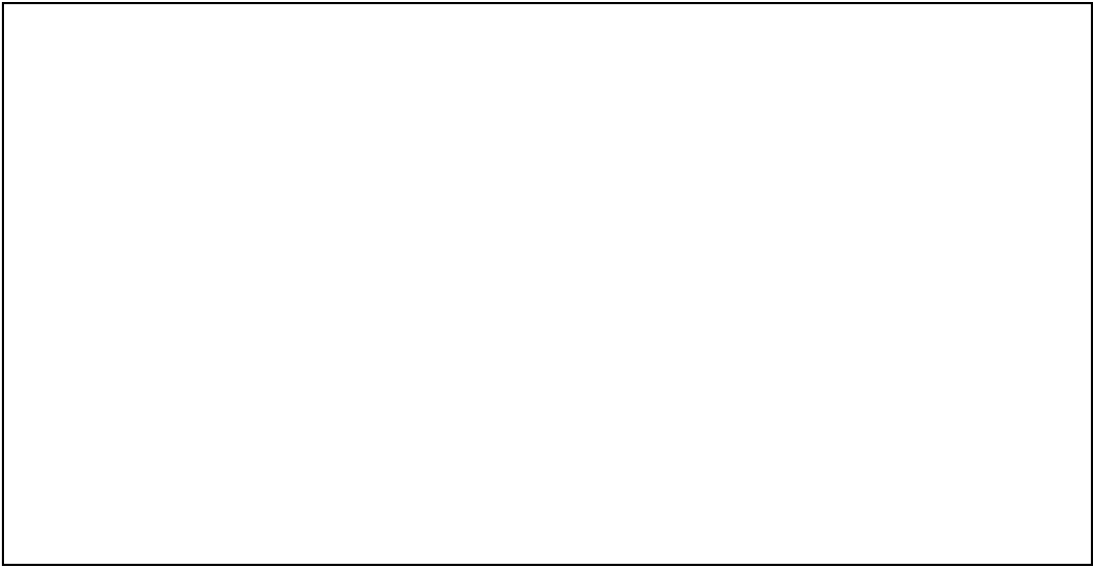
▶ $U_L = V_t \frac{R_L}{R_t + R_L}$

▶ $I_L = \frac{V_t}{R_t + R_L}$

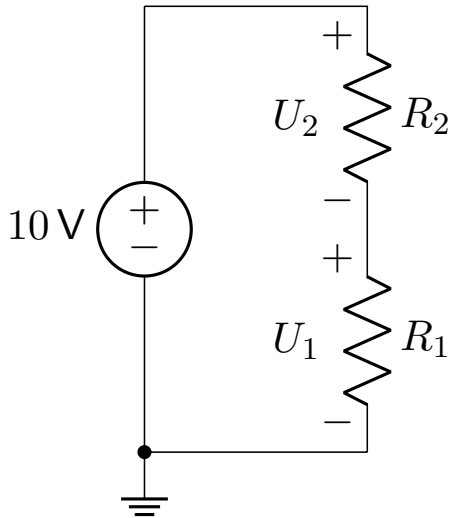
▶ $P_L = V_t^2 \frac{R_L}{(R_t + R_L)^2}$

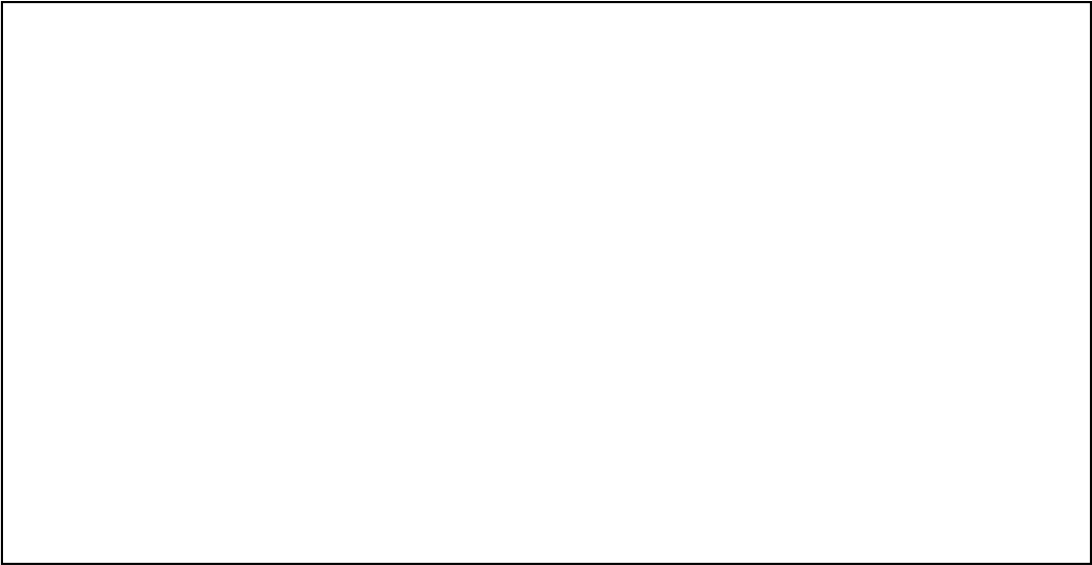
▶ P_L störst då $R_L = R_t$

$$P_{L\max} = \frac{V_t^2}{4R_t}$$

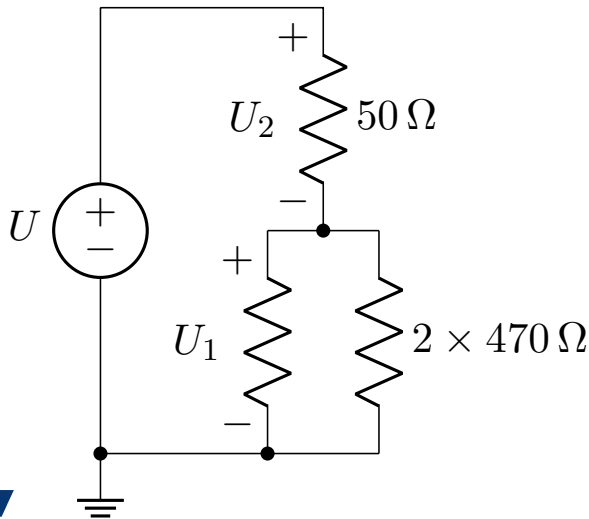


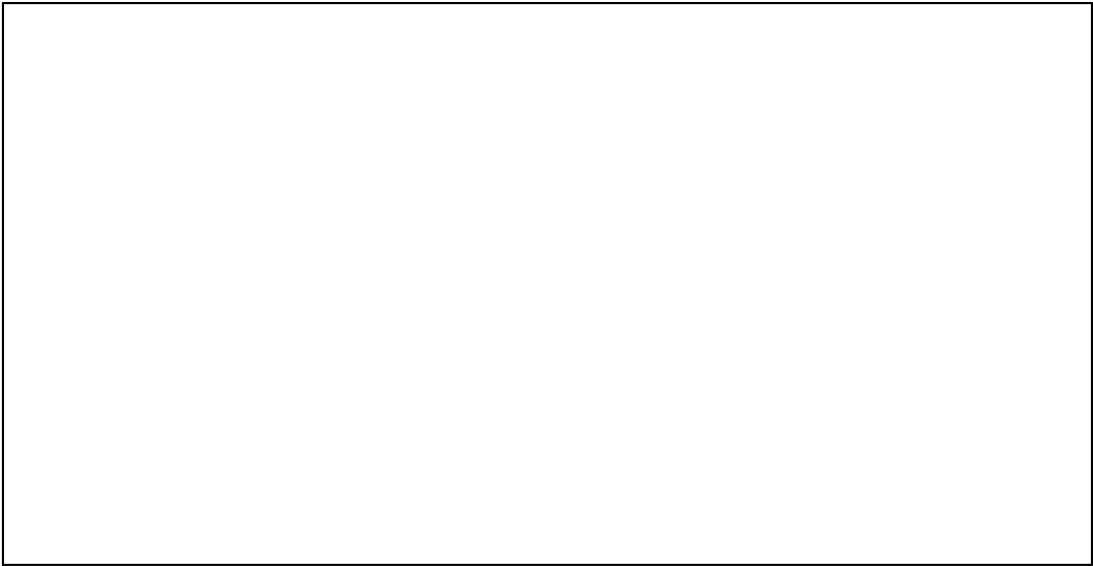
$R_1 = 68 \text{ k}\Omega$ och $R_2 = 10 \text{ k}\Omega$. Beräkna U_1 och U_2 .



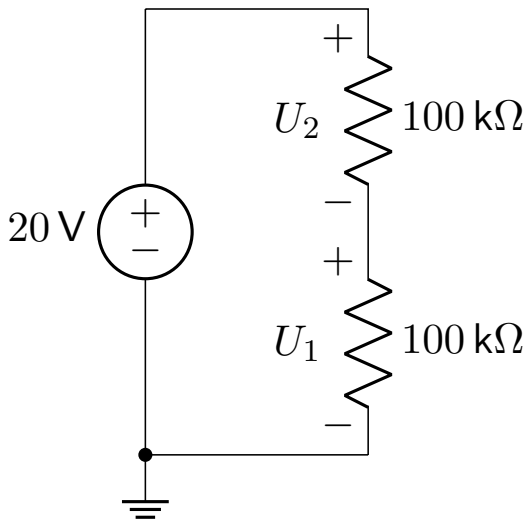


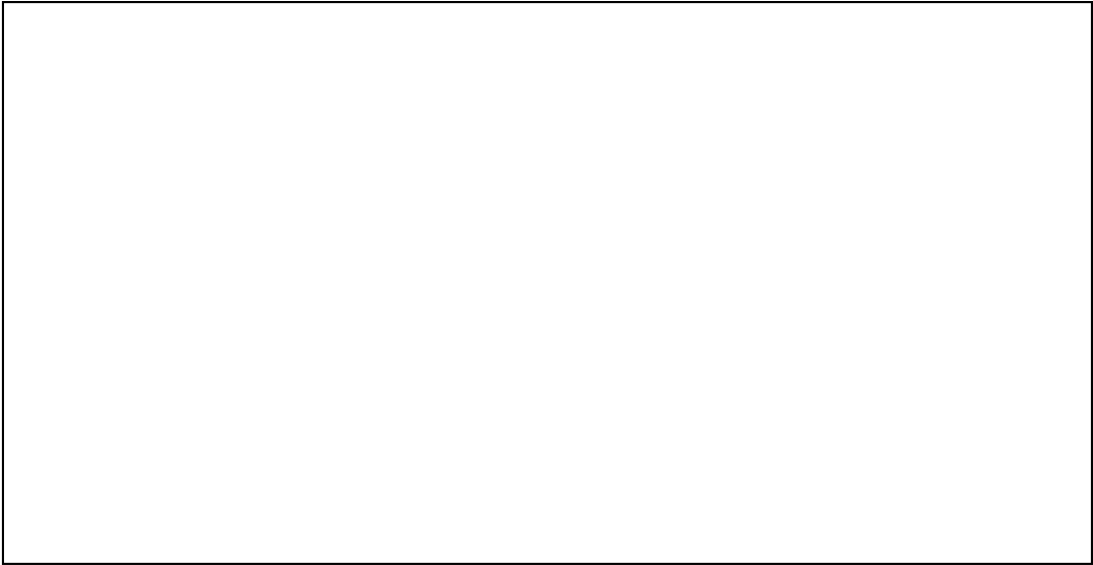
Koppla upp kretsen nedan. Använd samma ström för $50\,\Omega$ -motståndet som igår, vad skall då ställas in för U ? Kontrollmät motstånden innan inkoppling. Mät U_1 och U_2 och säkerställ att $U = U_1 + U_2$.



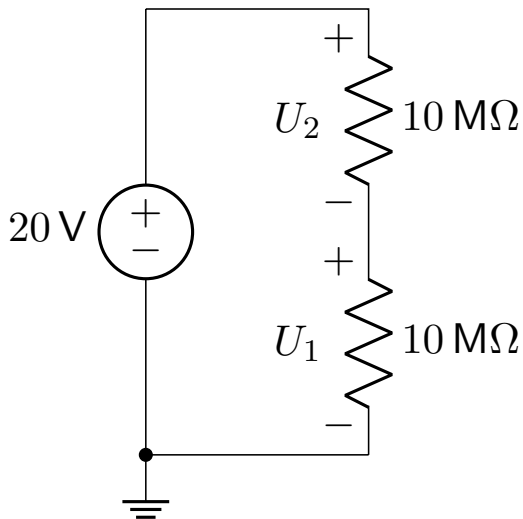


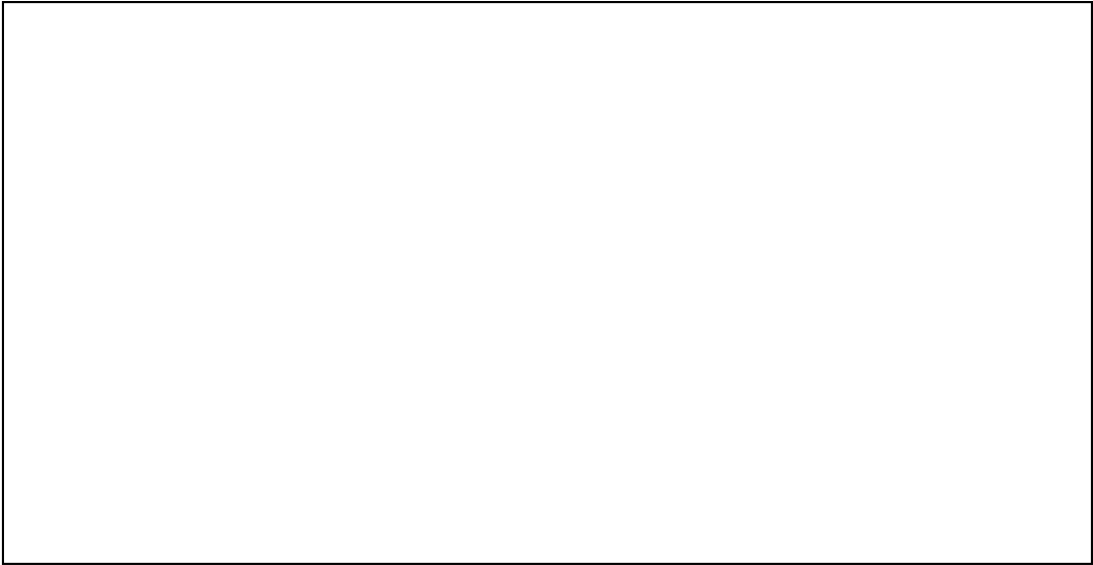
Koppla upp kretsen nedan. Kontrollmät motstånden innan inkoppling. Mät U_1 och U_2 och säkerställ att $U = U_1 + U_2$.





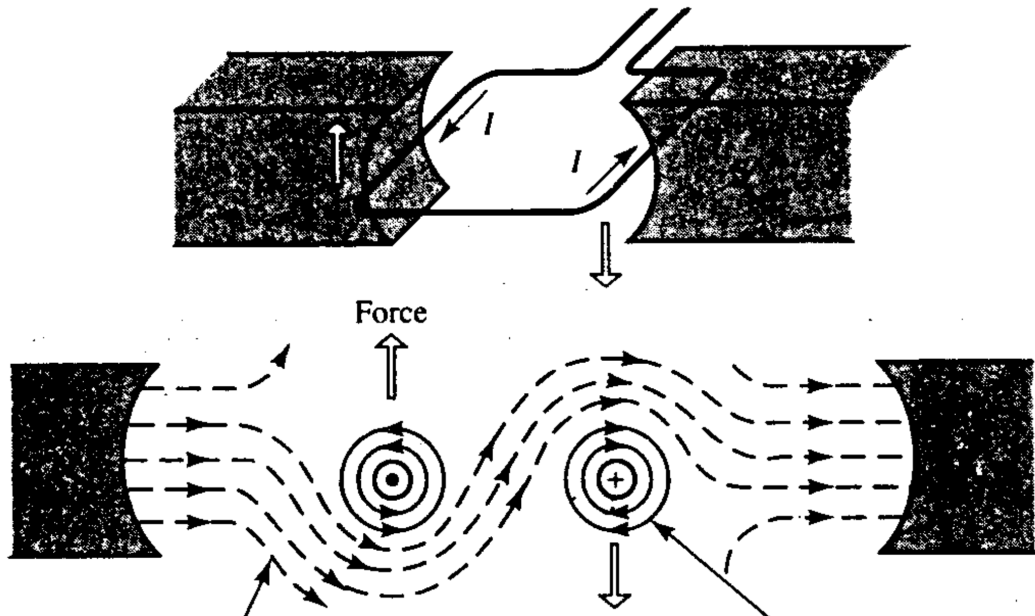
Koppla upp kretsen nedan. Kontrollmät motstånden innan inkoppling. Mät U_1 och U_2 och säkerställ att $U = U_1 + U_2$.



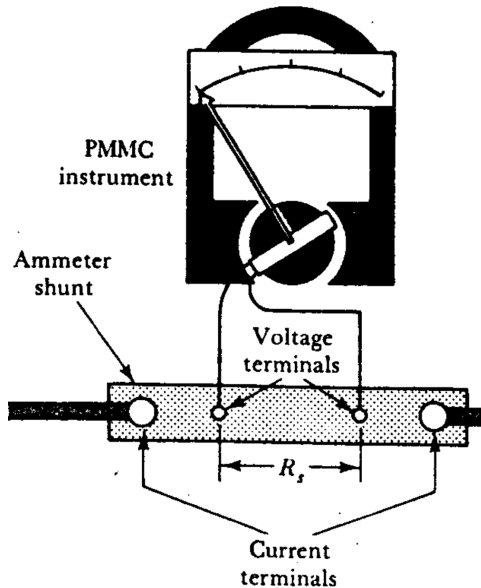
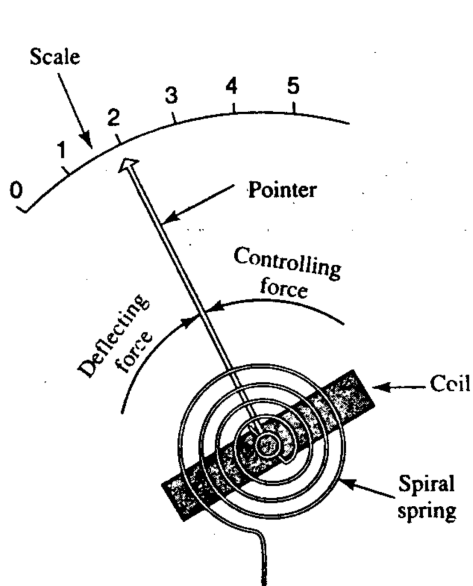


Ett mätinstrument drar ström

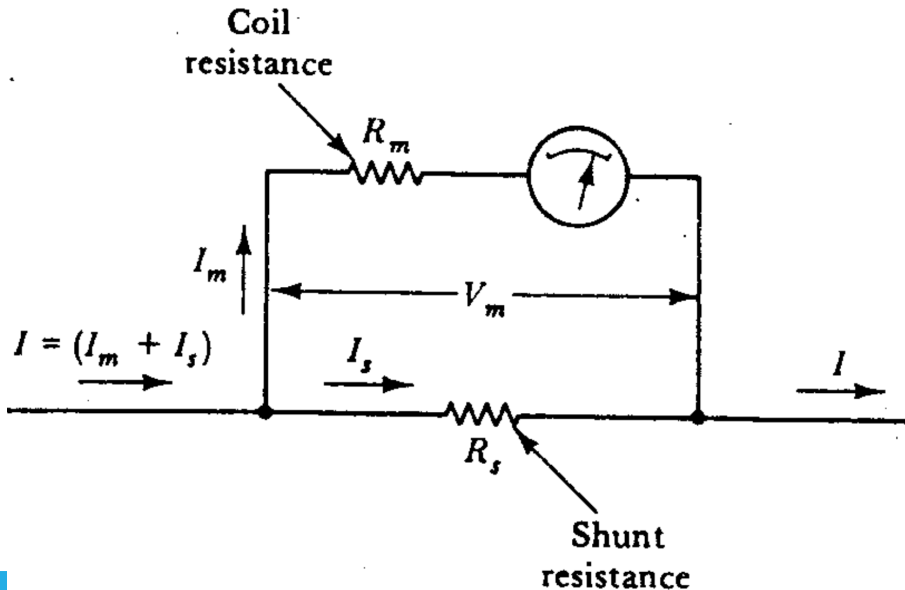
Så funkar ett vridspoleinstrument – principen



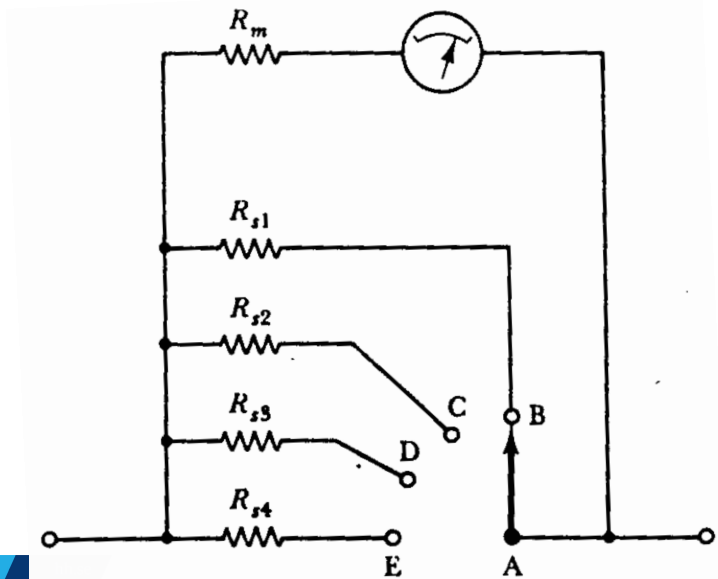
Så funkar ett vridspoleinstrument – ingenjörskonsten



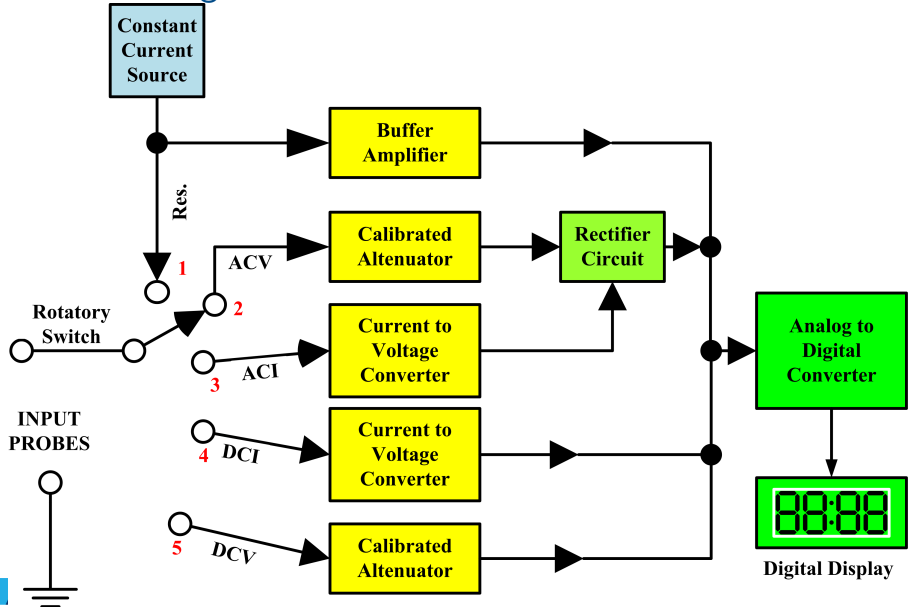
Så funkar ett vridspoleinstrument – kretsen



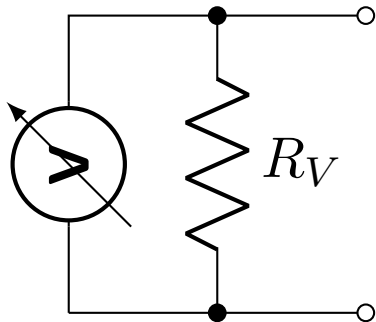
Så funkar ett vridspoleinstrument – flera mätområden



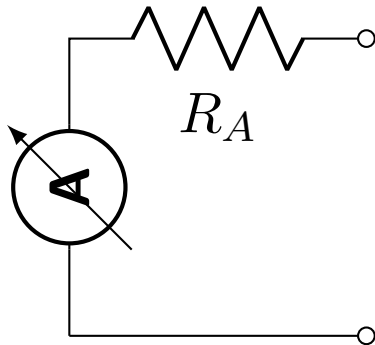
Så funkar ett digitalt instrument



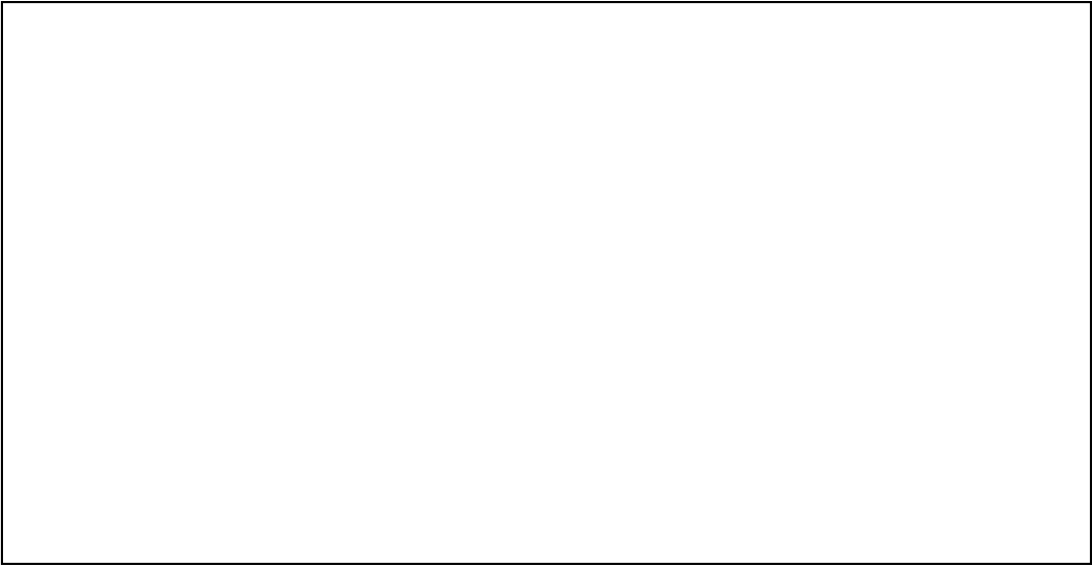
Inre resistans i instrument!



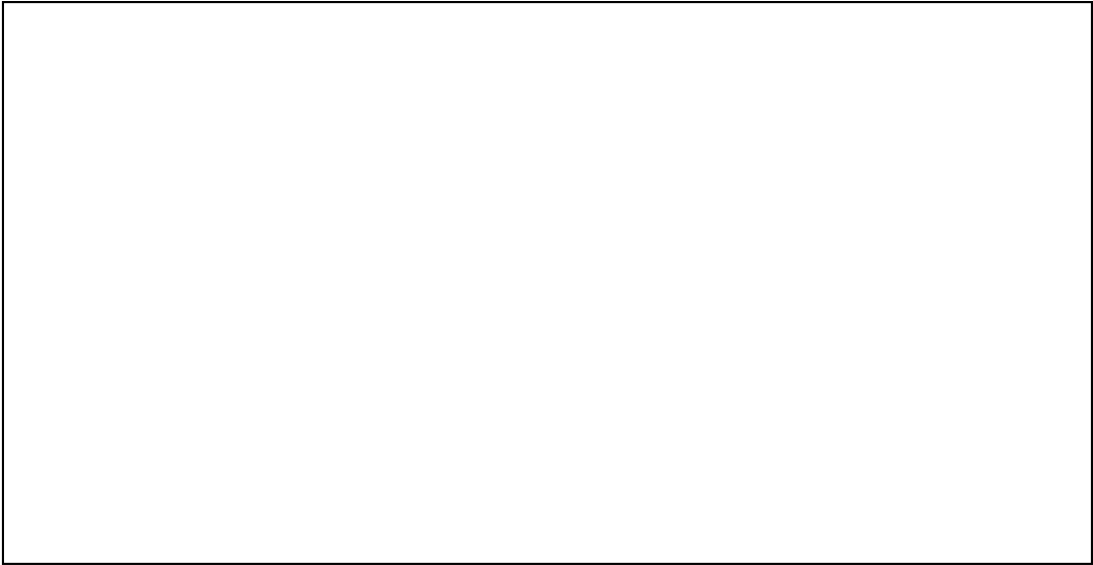
Voltmeter



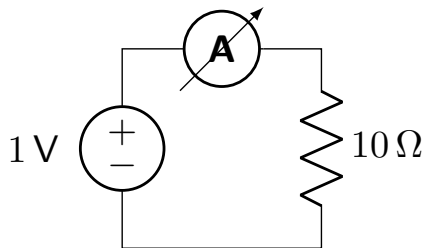
Amperemeter



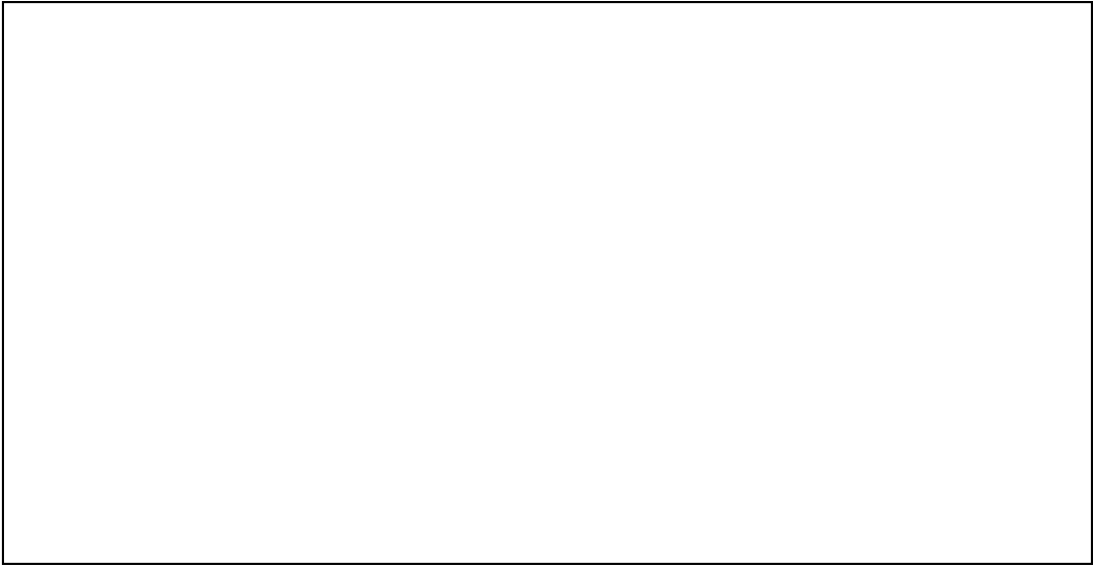
- ▶ Mät inre resistansen i ert instrument!
- ▶ Räkna (med spänningsdelning!) vad just din multimeter borde visat vid mätning av $10\text{ M}\Omega$ -delningen.



Inre resistans – amperemätning



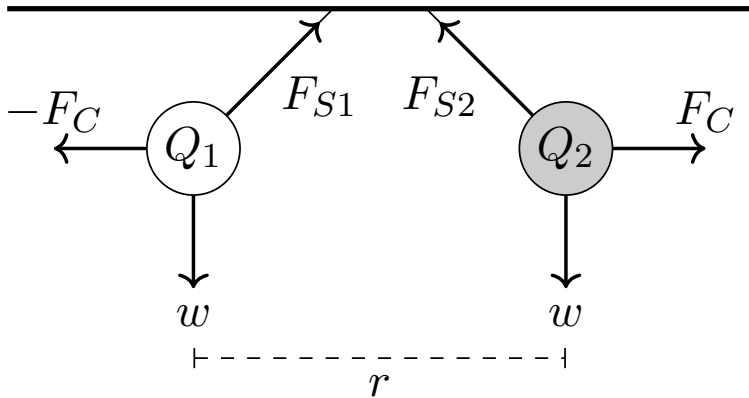
- ▶ Mät ström mA-ingång.
- ▶ Koppla i och ur amperemetern.
- ▶ Betrakta strömmen på *spänningskuben*.
- ▶ Beräkna R_A .



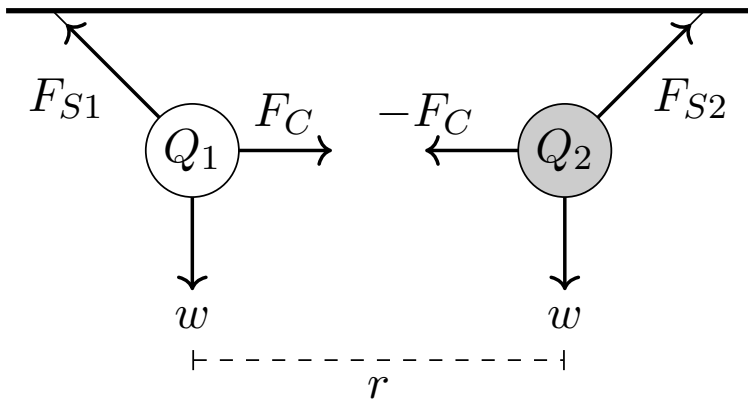
Lite om statisk elektricitet



Lika laddningar repellerar



Olika laddningar attraherar



Kondensatorn

- ▶ Sparar energi $\Rightarrow$ omfördela laddningar på **kondensatorn**
- ▶ Kondensatorns **kapacitans** def:

$$C = \frac{Q}{U} \Rightarrow Q = C \cdot U$$

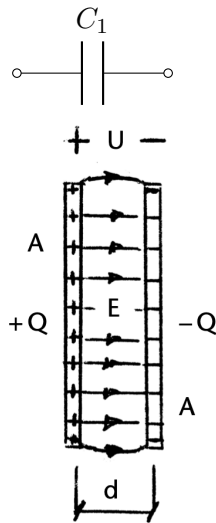
(enhet Farad – $1 \text{ F} = 1 \text{ kg}^{-1} \text{ m}^{-2} \text{ s}^4 \text{ A}^2$)

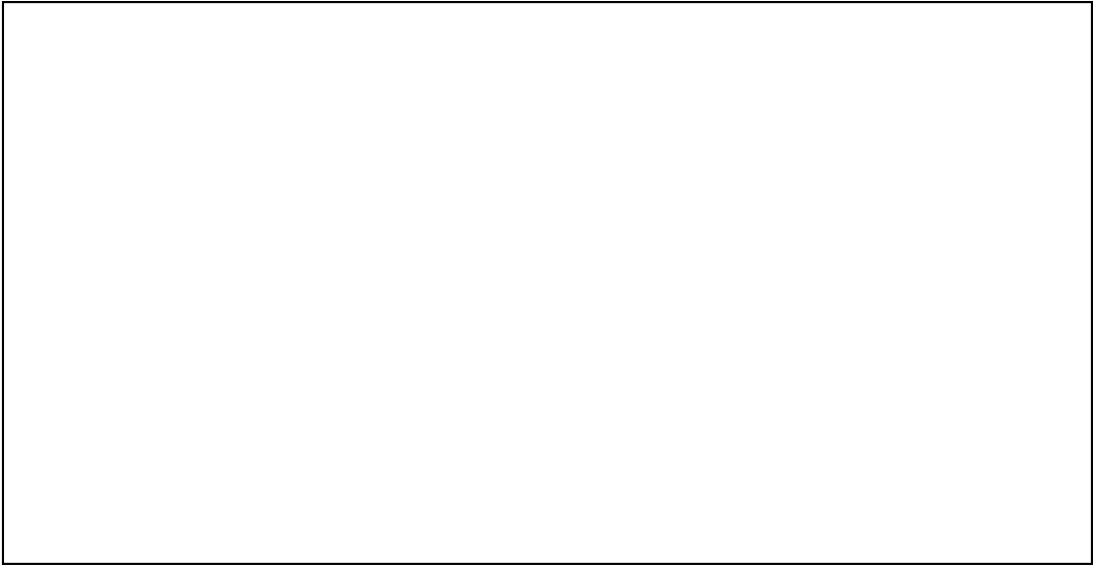
- ▶ Större ytor $\Rightarrow$ fler laddningar
- ▶ Tätare plattor $\Rightarrow$ laddningar attraheras

$$C = \frac{A \epsilon_0 \epsilon_r}{d}$$

Luft mellan plattorna $\epsilon_r = 1$

Material	ϵ_r
Luft, torr	1,0005
Glimmer	≈ 6
Polykarbonat	2,9
Polypropylen	2,3
Teflon	2,1





Kapacitans – exempel

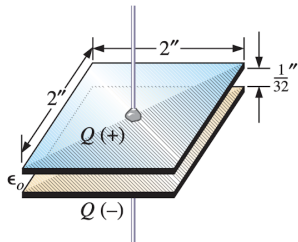


FIG. 10.10

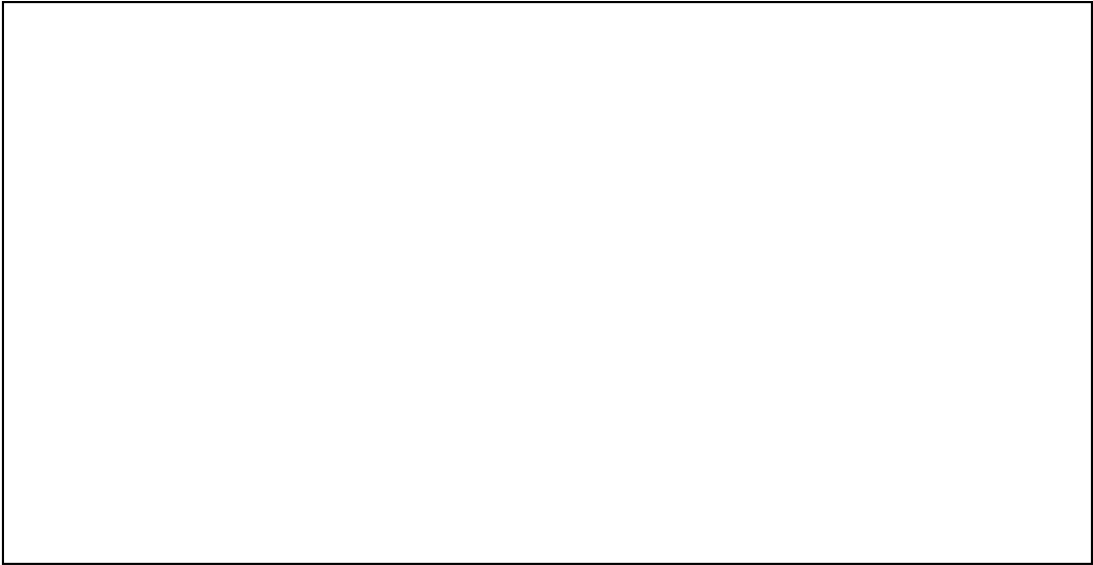
Air capacitor for Example 10.3.

$$C = \frac{A\epsilon_0\epsilon_r}{d}$$

$$C = \frac{Q}{U} \Rightarrow Q = C \cdot U$$

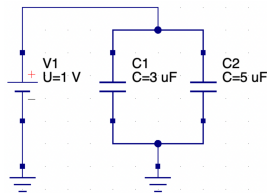
EXAMPLE 10.3 For the capacitor in Fig. 10.10:

- Find the capacitance.
- Find the strength of the electric field between the plates if 48 V are applied across the plates.
- Find the charge on each plate.



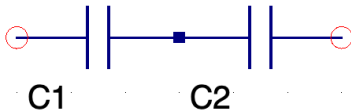
Kondensatorer i kretsar

- ▶ Batteri laddar upp kondensator
- ▶ **OBS** elektrolytkond **rätt håll!**



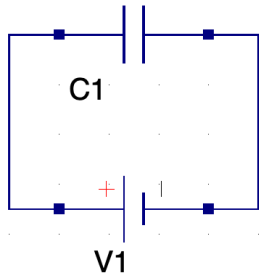
Parallella $\Rightarrow$ högre kapacitans!

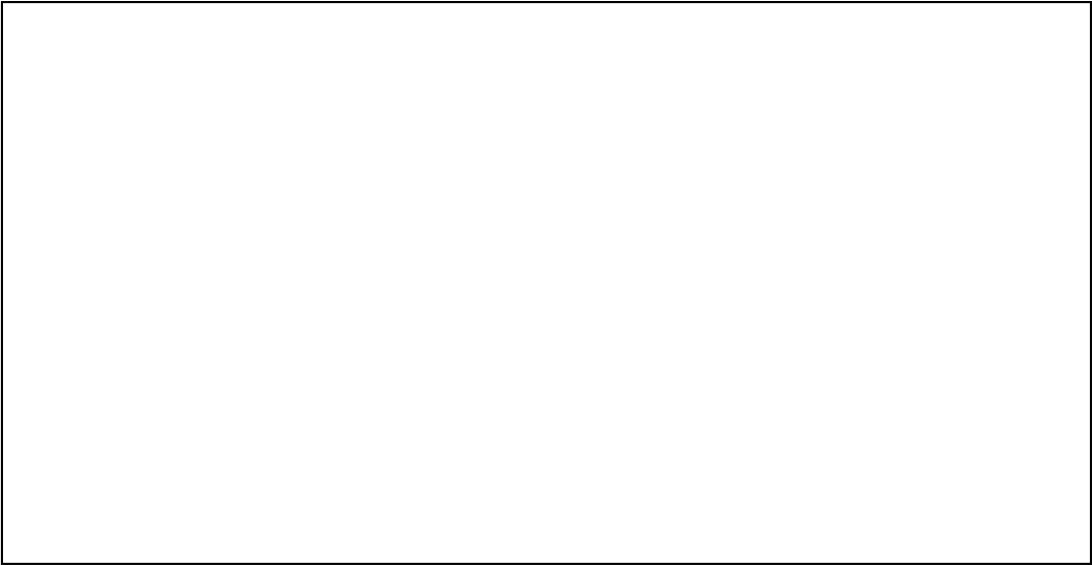
$$C_{\text{tot}} = C3 + C4 + \dots$$



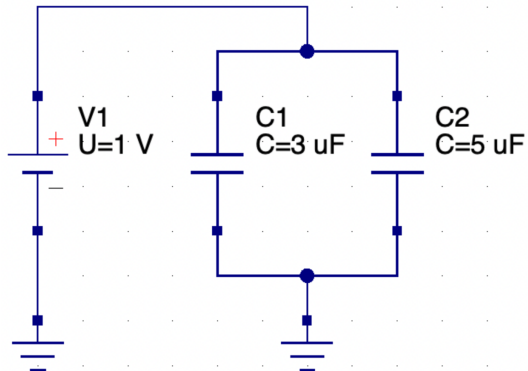
Serie $\Rightarrow$ lägre kapacitans!

$$\frac{1}{C_{\text{tot}}} = \frac{1}{C1} + \frac{1}{C2} + \dots$$



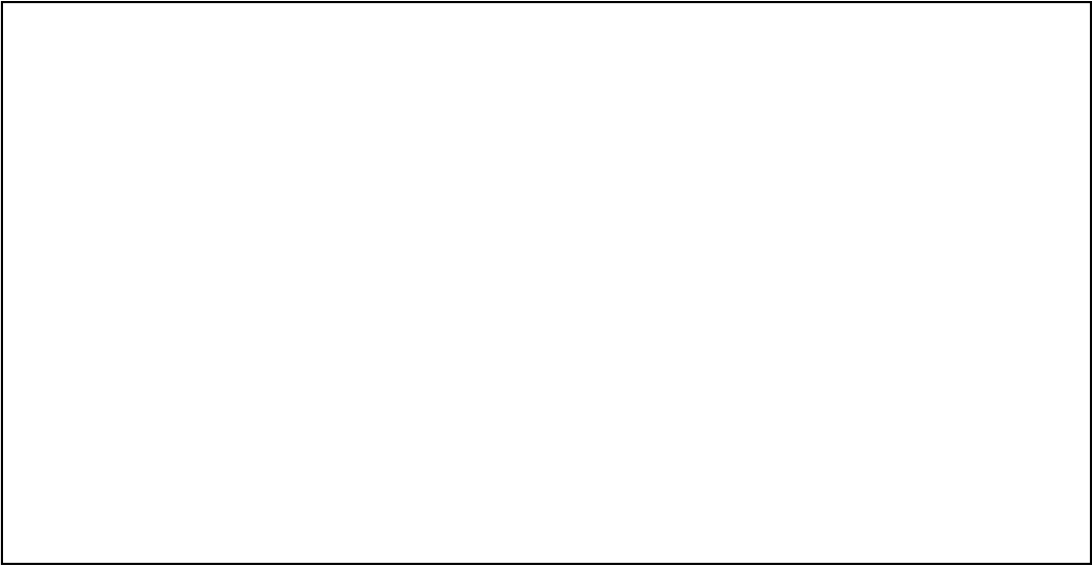


Kondensatorer i kretsar



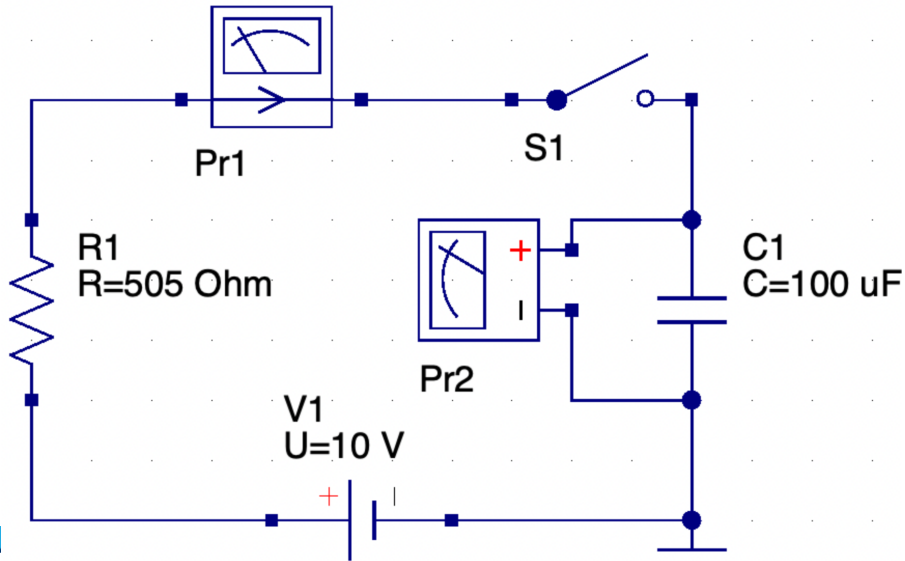
- ▶ Bestäm laddningen på kondensatorerna.
- ▶ Hur stor ström går det i kretsen efter lång tid?

$$C = \frac{Q}{U} \Rightarrow Q = CU$$



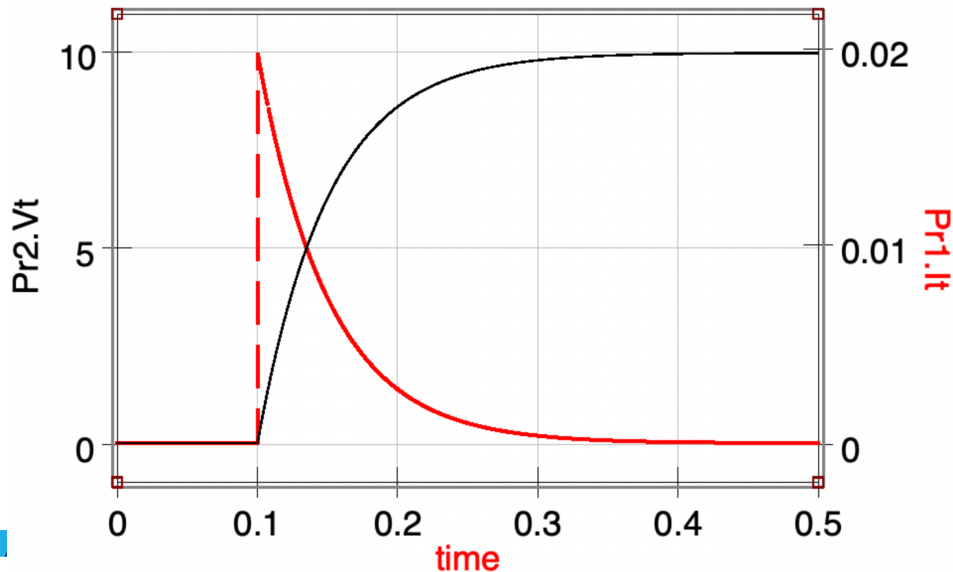
Uppladdning av kondensator

OBS! Alltid kondensatorer i serie med resistor



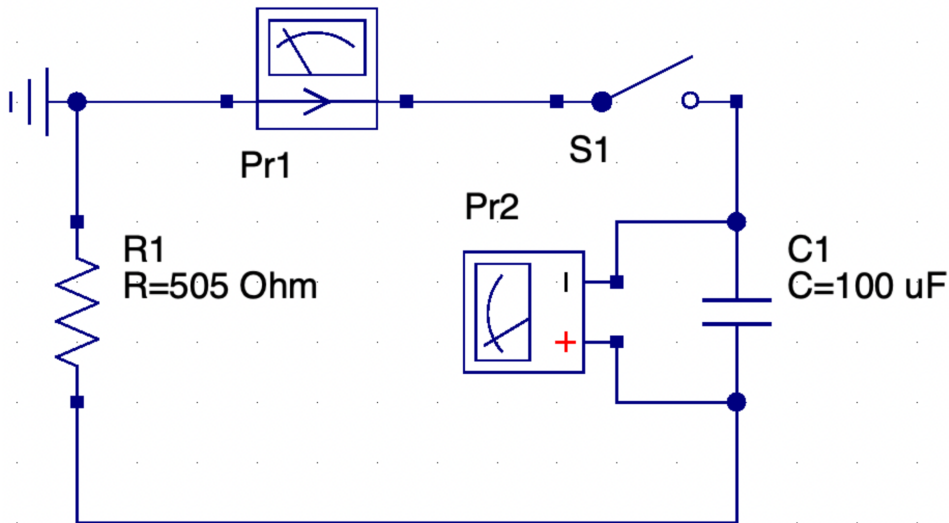
Uppladdning av kondensator

OBS! Alltid kondensatorer i serie med resistor



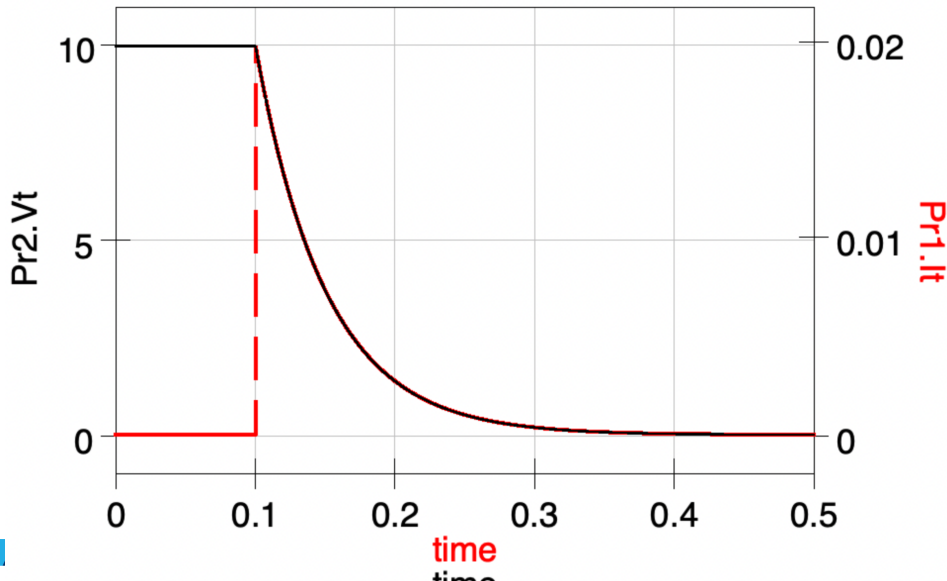
Urladdning av kondensator

OBS! Alltid kondensatorer i serie med resistor



Urladdning av kondensator

OBS! Alltid kondensatorer i serie med resistor



Upp- och urladdning av kondensator

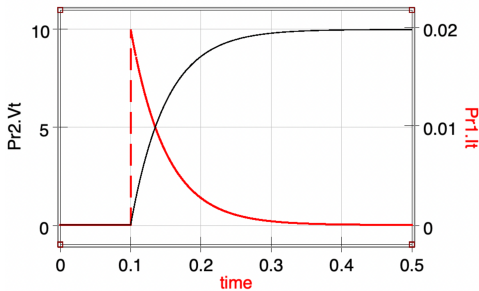
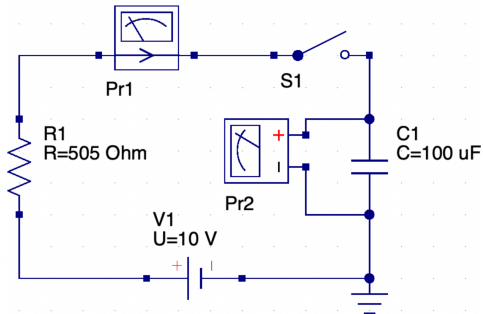
OBS! Alltid kondensatorer i serie med resistor Spänning i kretsen beskrivs av:

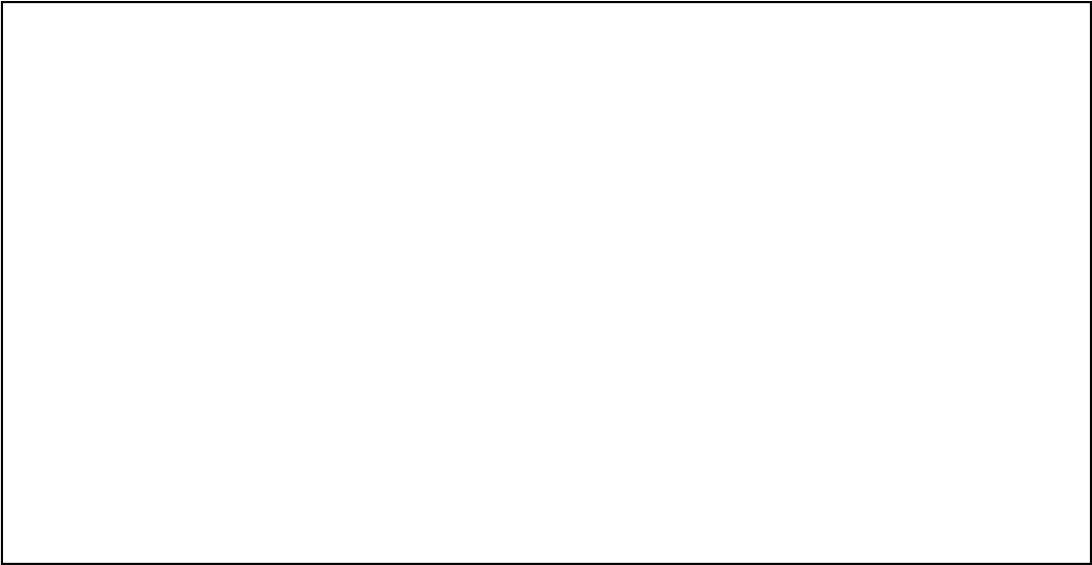
$$\frac{du}{dt} + \frac{1}{RC}u = 0.$$

kretsens tidskonstant:

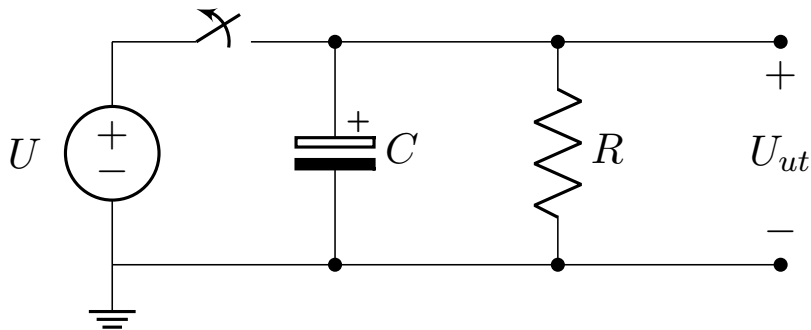
$$\tau = RC$$

Efter tiden τ 63% av ändringen.





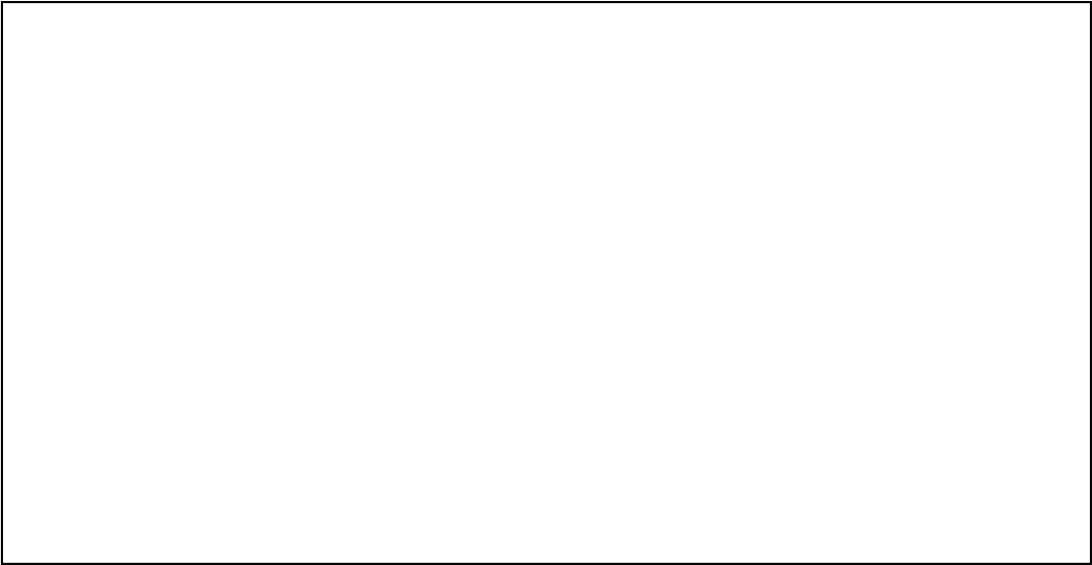
Urladdning av kondensator – uppkoppling

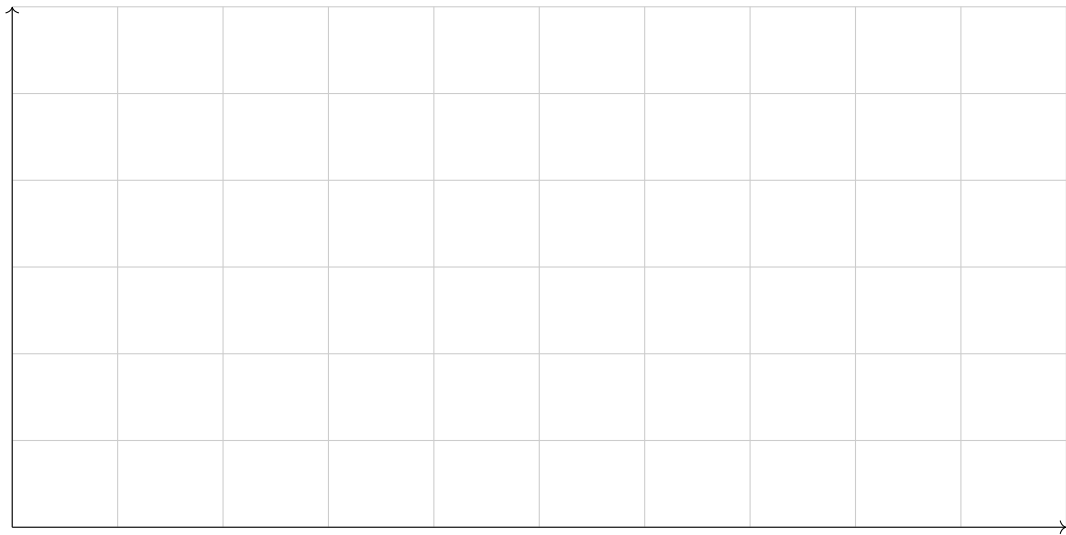


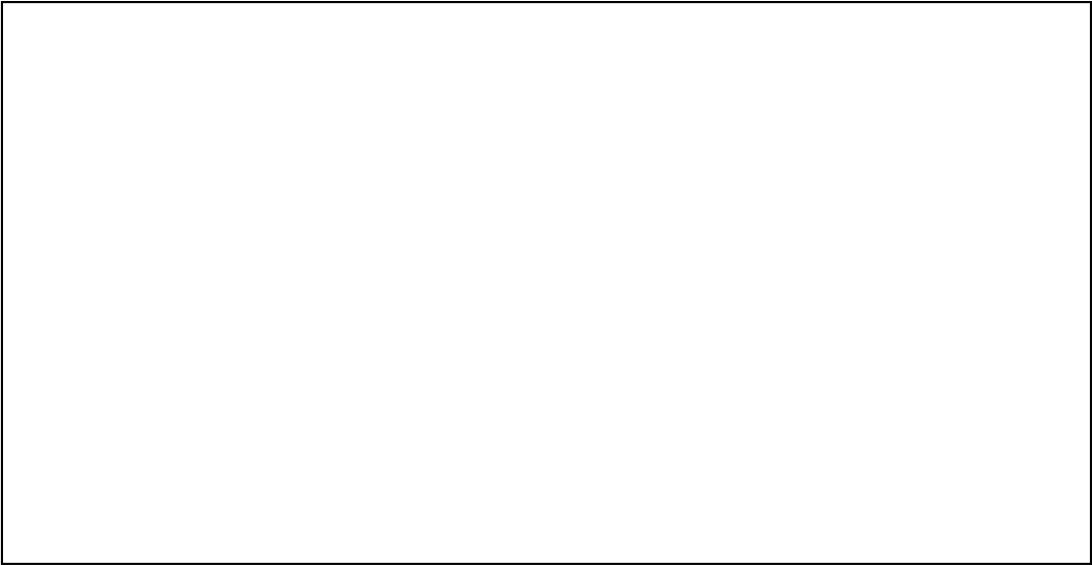
$$U < U_{\max}. \quad \tau = RC \approx 1 \text{ min.}$$

Nedteckna U_{ut} minst 15 ggr tills $t > 1.5\tau$.

Presentera **tydliga** grafer!

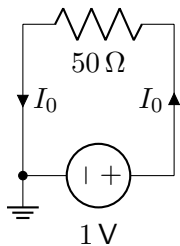






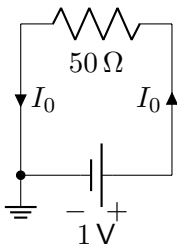
Direct Current och Alternating Current

DC – Likström

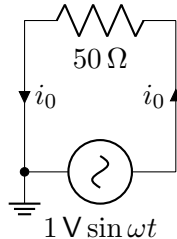


Ohms lag ger:

$$I_0 = \frac{1\text{ V}}{50\ \Omega} = 20\text{ mA}$$

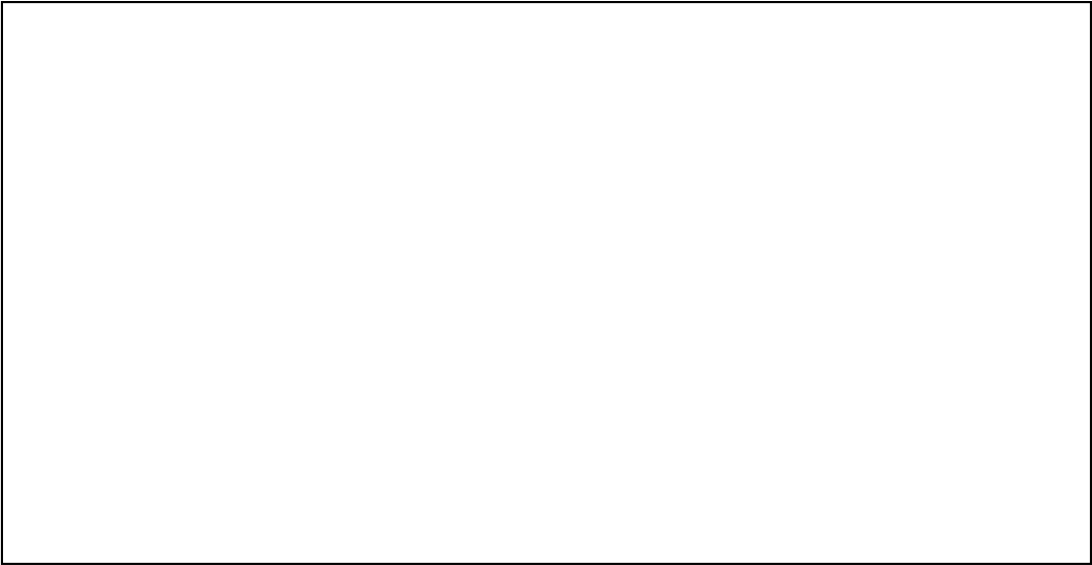


AC – Växelström

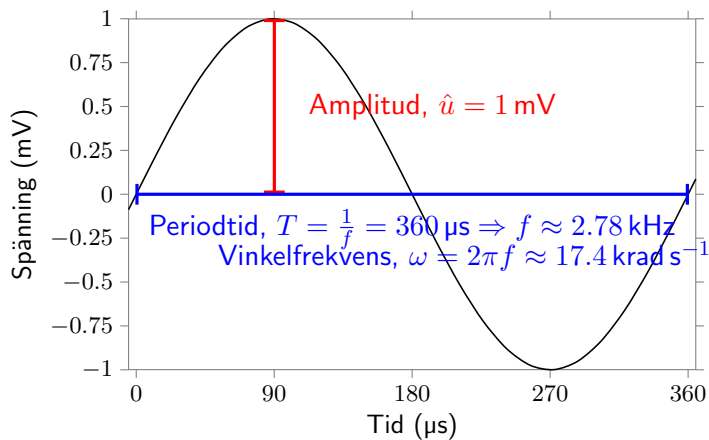


Ohms lag ger (också):

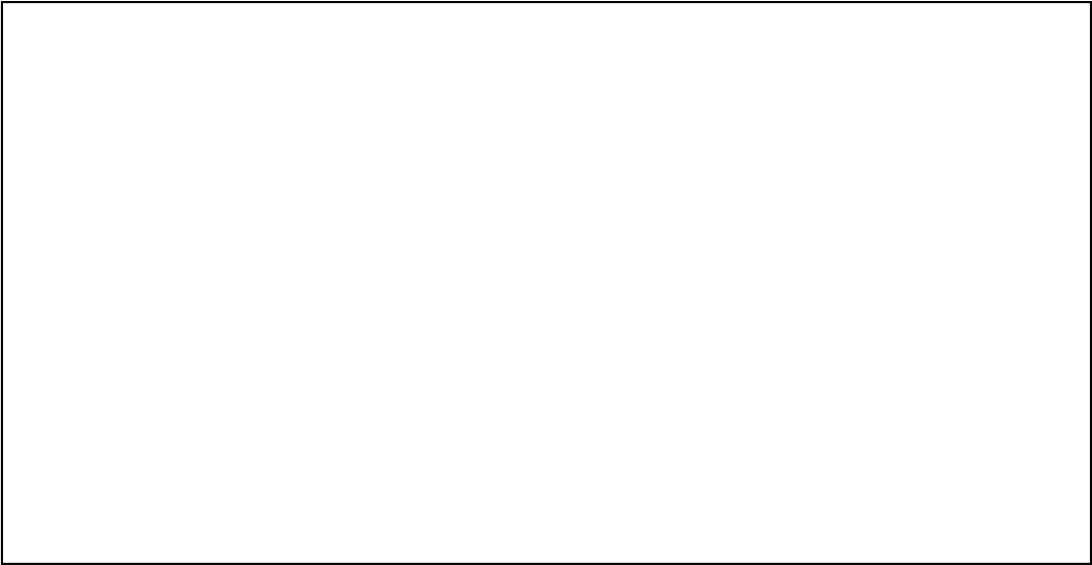
$$i_0 = \frac{1\text{ V sin } \omega t}{50\ \Omega} = 20\text{ mA sin } \omega t$$



Växelström och växelspanning



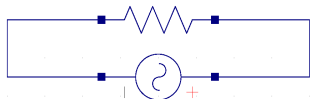
$$u(t) = 1 \text{ mV} \sin(17.4 \text{ krad s}^{-1} t)$$



Växelström - Resistorn

Växelspänning

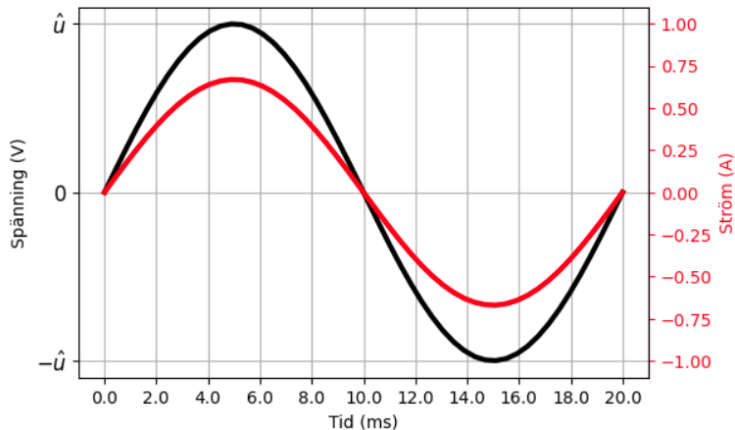
$$u(t) = \hat{u} \sin \omega t$$

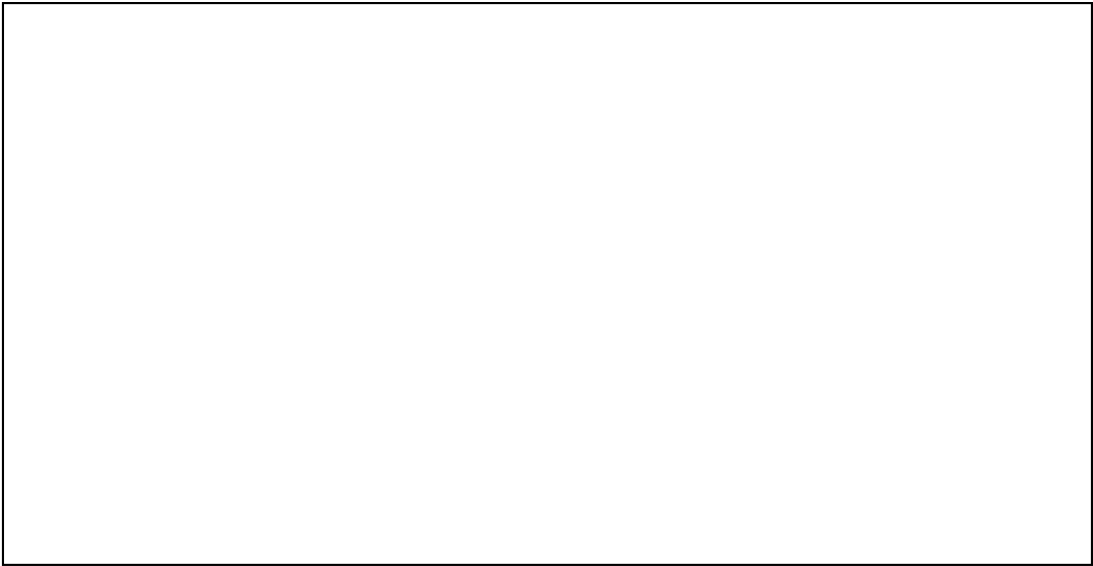


Ohms vanliga lag gäller:

$$u = R \cdot i$$

både amplituder och
effektivvärden.

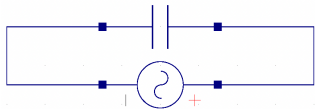




Växelström - Kondensatorn

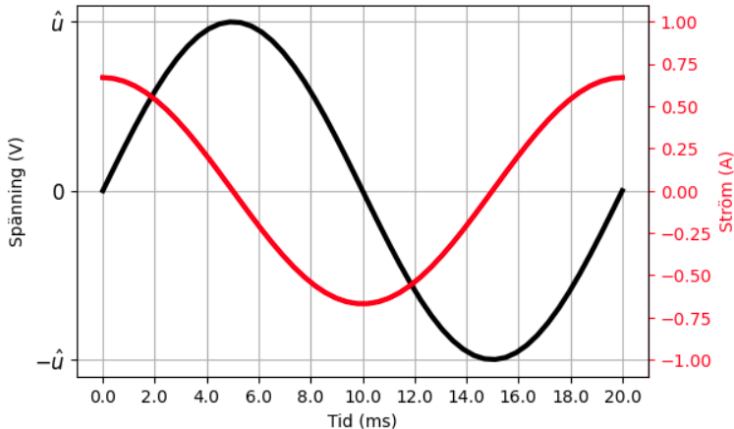
Växelspänning

$$u(t) = \hat{u} \sin \omega t$$

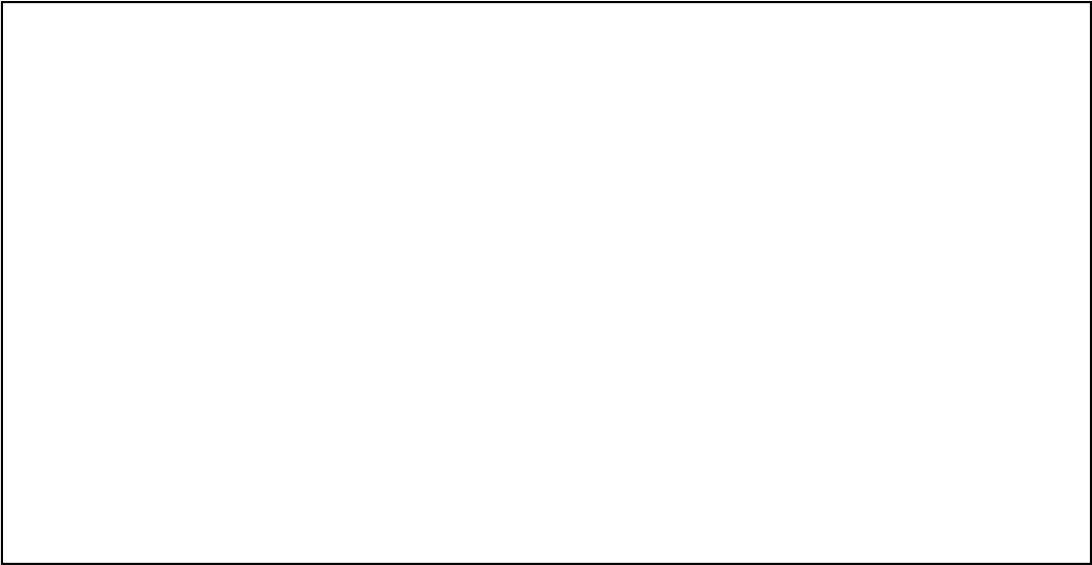


Strömmen går 90° före spänningen.

Ohms lag $U = \frac{1}{\omega C} I$
effektivvärden och
amplituder.



$\frac{1}{\omega C}$ kapacitiva reaktansen $X_C (\Omega)$



Över en kondensator med kapacitansen $0.1 \mu\text{F}$ ligger spänningen $u(t) = 100 \text{ V} \sin 400\pi t$. Beräkna amplituden hos strömmen genom kondensatorn.

